AI for Video

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Overview

Video

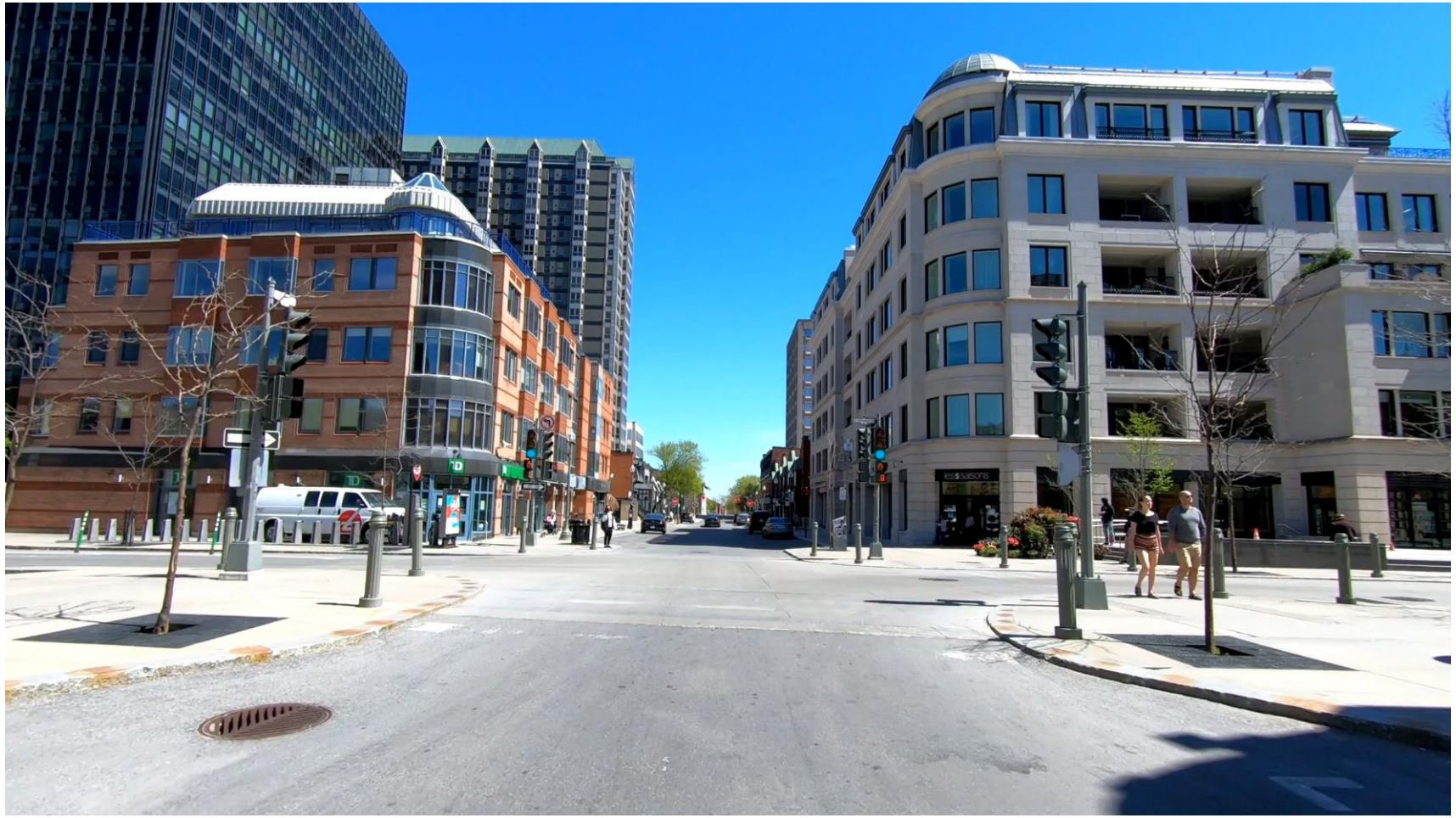
Video Analysis

Video Synthesis

Demo

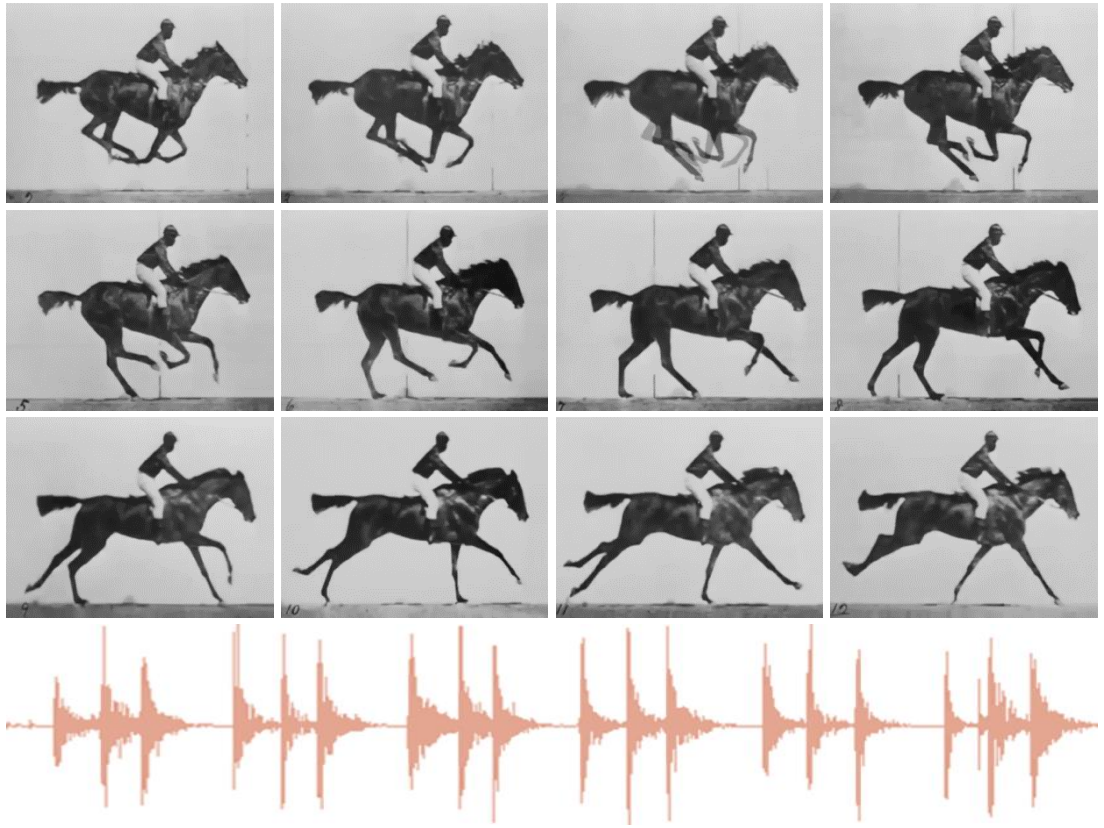
Video

Video



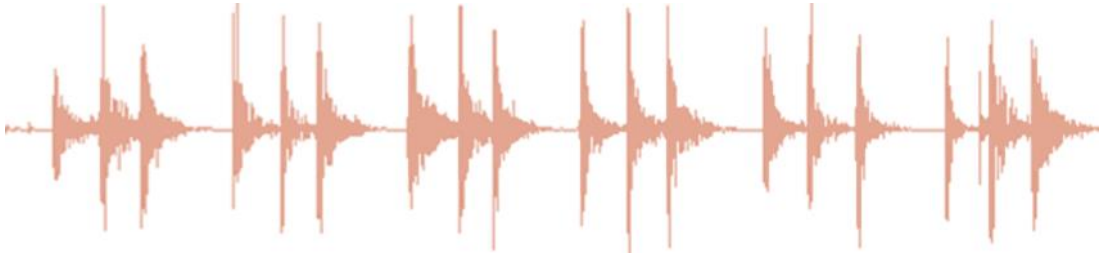
Video by German Korb from Pexels

Video

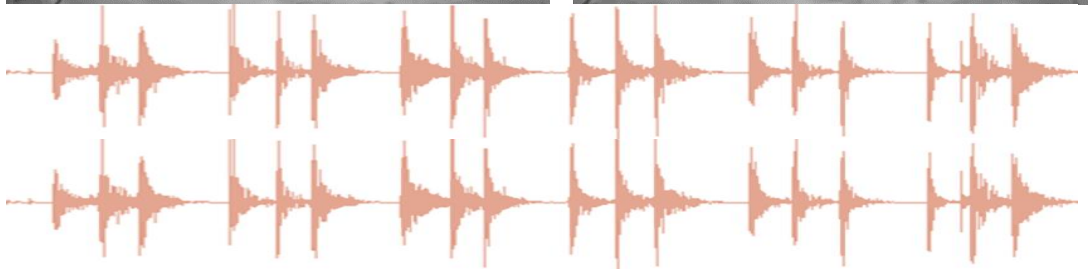


Video

Mono



Stereo



Includes All Previous A/V Tasks

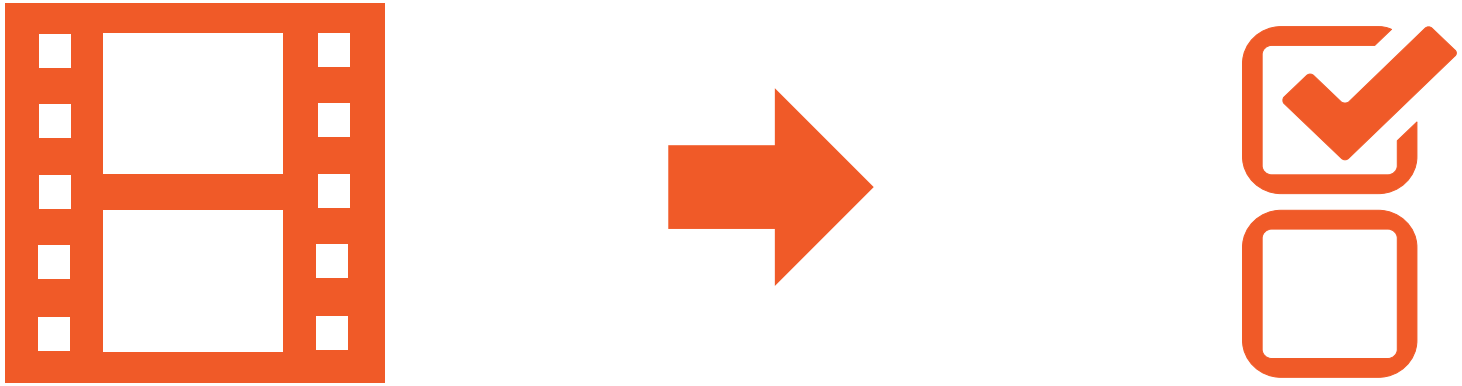
Audio Tasks	Image Tasks
Sound classification	Image classification
Voice recognition	Object detection
Speech recognition	Face recognition

Video Analysis

Video Analysis Tools

Motion detection	Object tracking	Action recognition

Motion Detection



Motion Detection

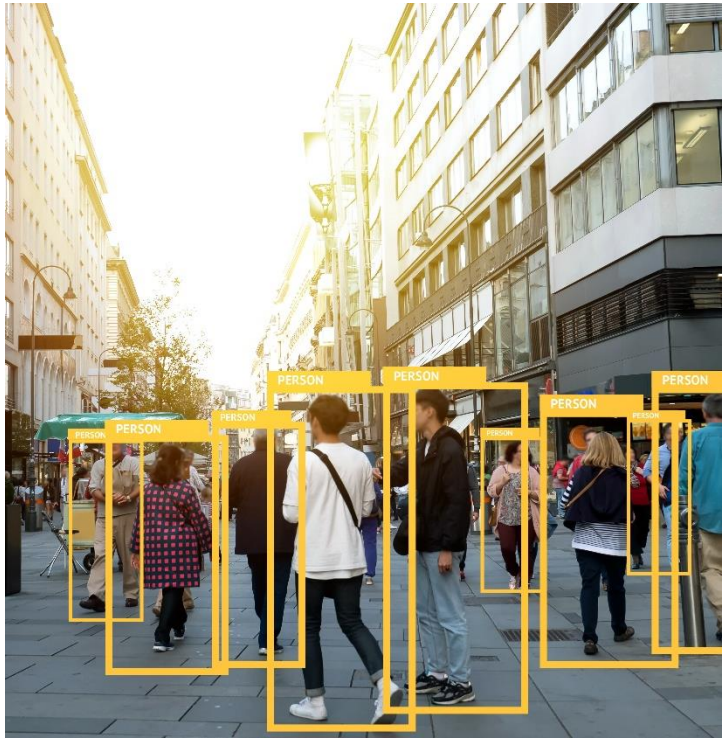
Input

Output

No Motion 100%

Motion

Motion Detection

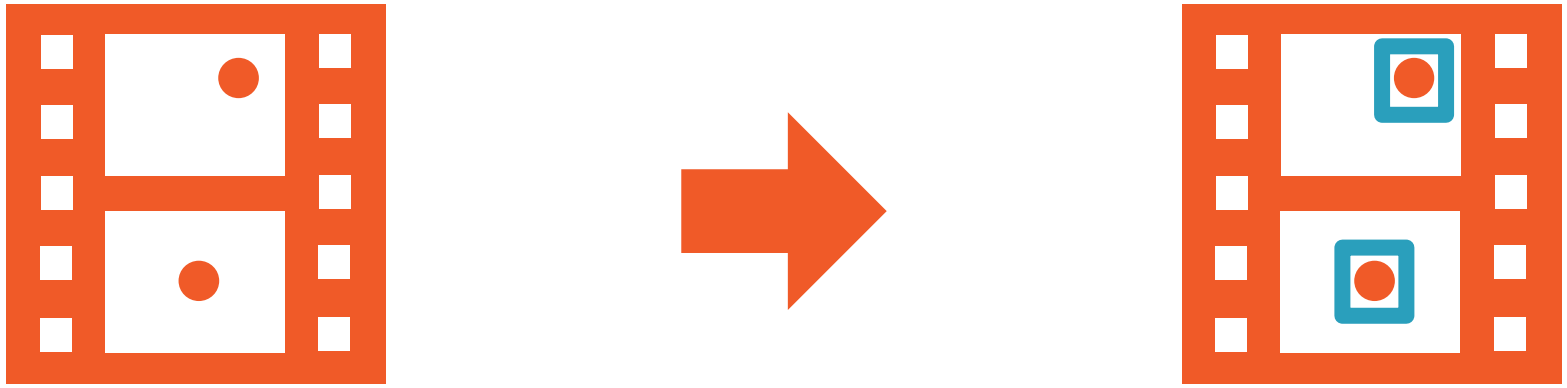


Video surveillance

Wildlife monitoring

Collaborative robotics

Object Tracking



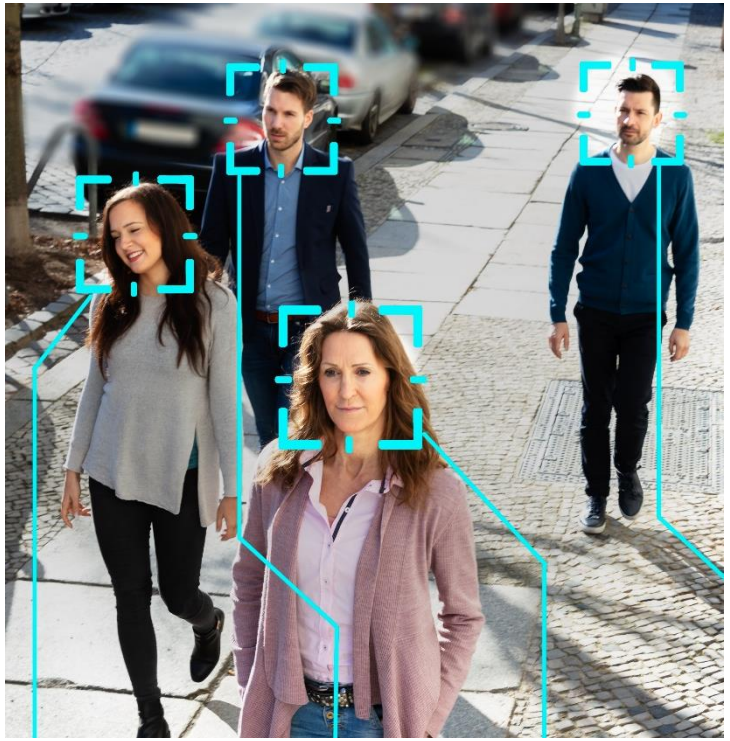
Object Tracking

Input

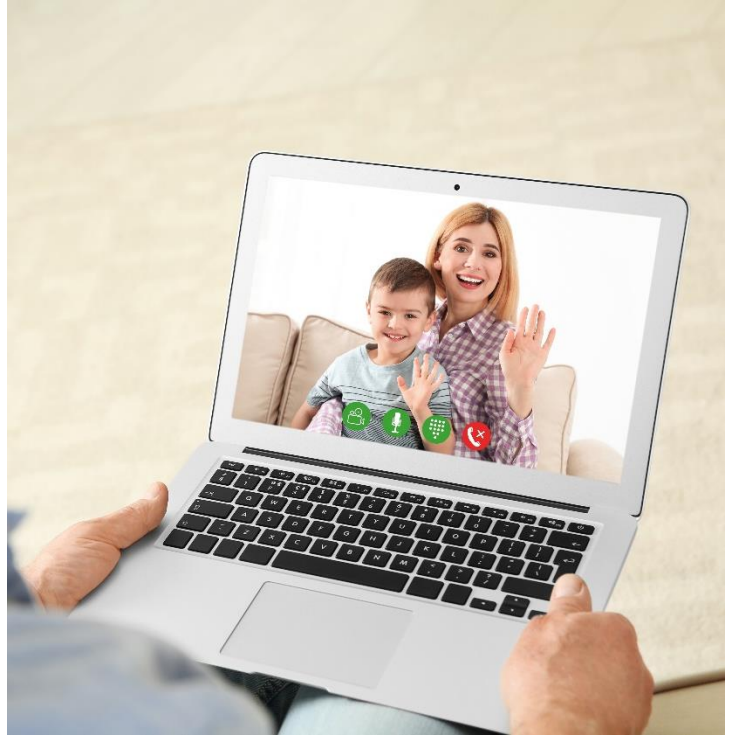
Output

Source: https://gluon-cv.mxnet.io/build/examples_tracking/demo_SiamRPN.html

Object Tracking



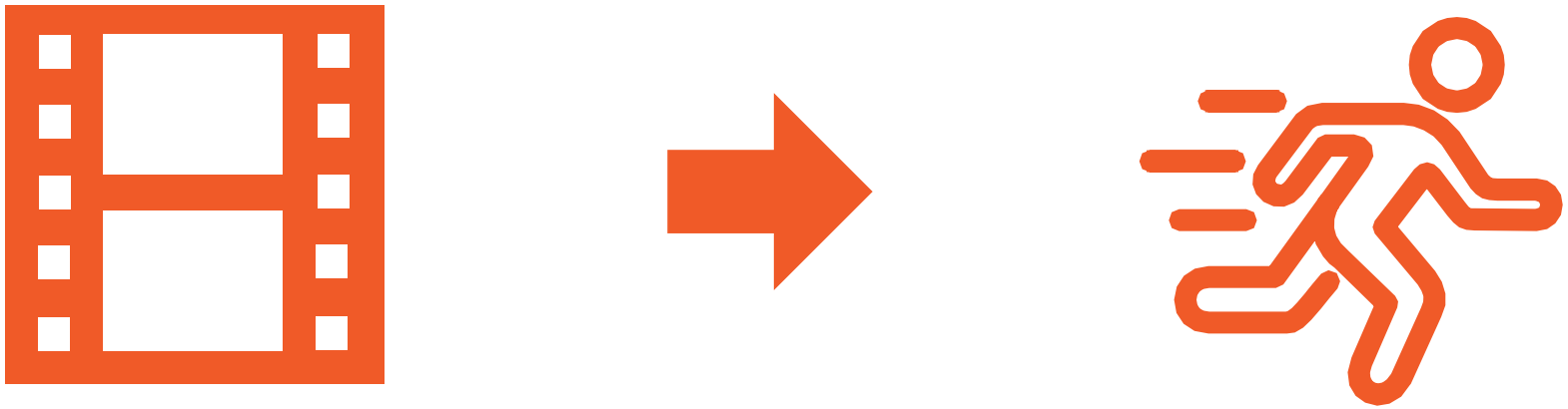
Video surveillance



Video conferencing

Self-driving vehicles

Action Recognition



Action Recognition

Input

Output

Walking

Standing

Sitting

Waving

Action Recognition

Video surveillance

Sports analysis

Robot collaboration

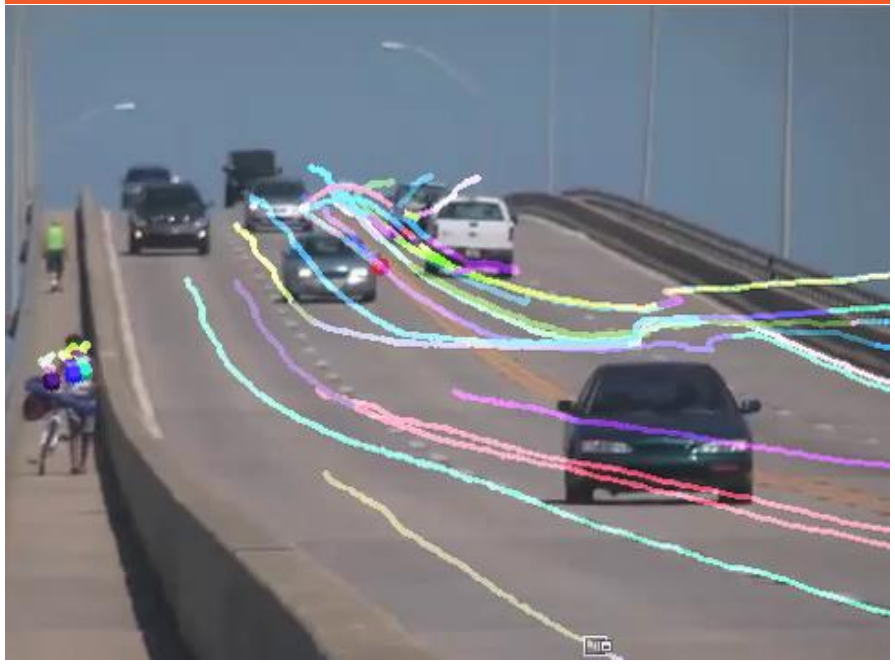
Other Video Analysis

Motion detection	Object tracking	Action recognition
Optical flow	Gait recognition	Lip reading

Optical Flow

Input

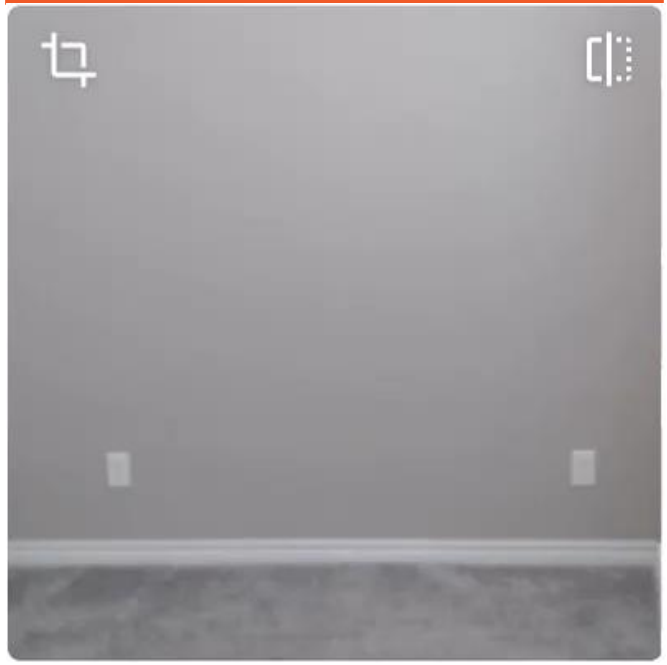
Output



Source: OpenCV

Gait Recognition

Input



Output

Person	Prediction
Matthew	
Zhang	
Priya	

Lip Reading

Input

Output

"We have to look at whether it works for the UK or not."

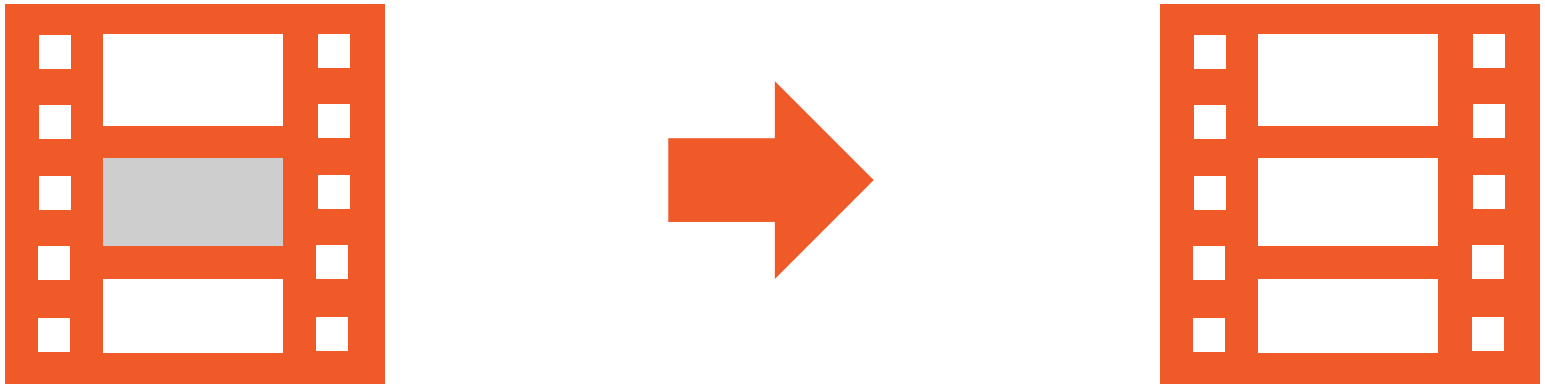
Source: Oxford, Lip Reading Sentences in the Wild, and BBC

Video Synthesis

Video Synthesis Tools

Video interpolation	**Video prediction**	**Video transfer**

Video Interpolation



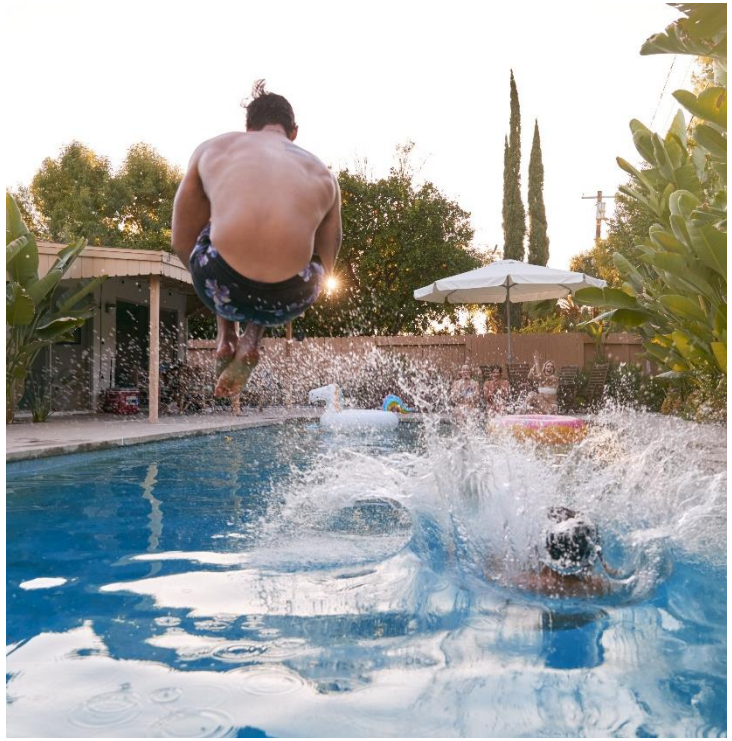
Video Interpolation

Input

Output

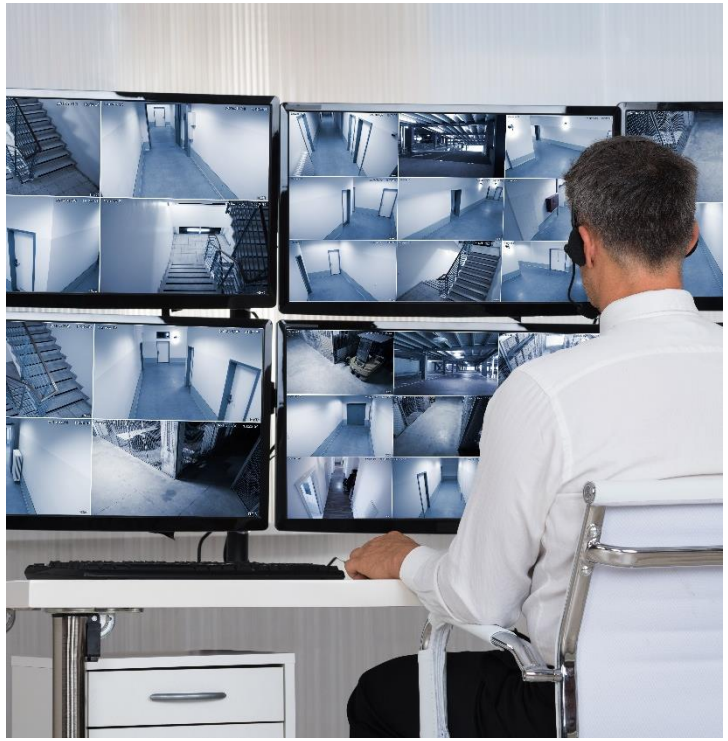
Source: Nvidia Super SloMo

Video Interpolation



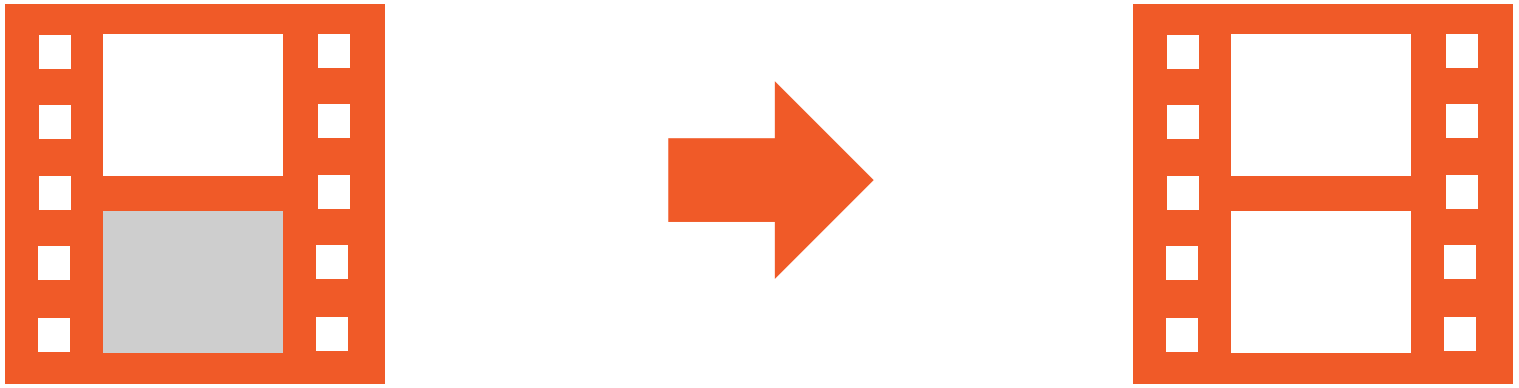
Frame-rate conversion

Film restoration



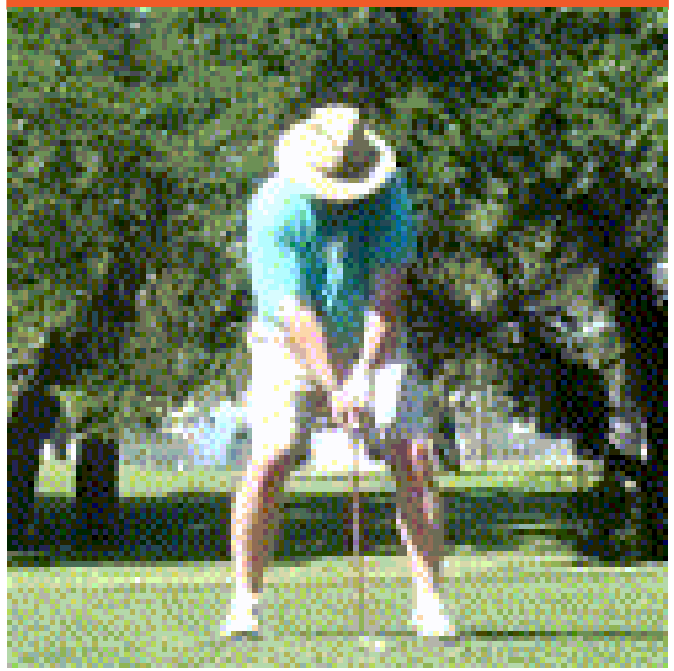
Video multiplexing

Video Prediction

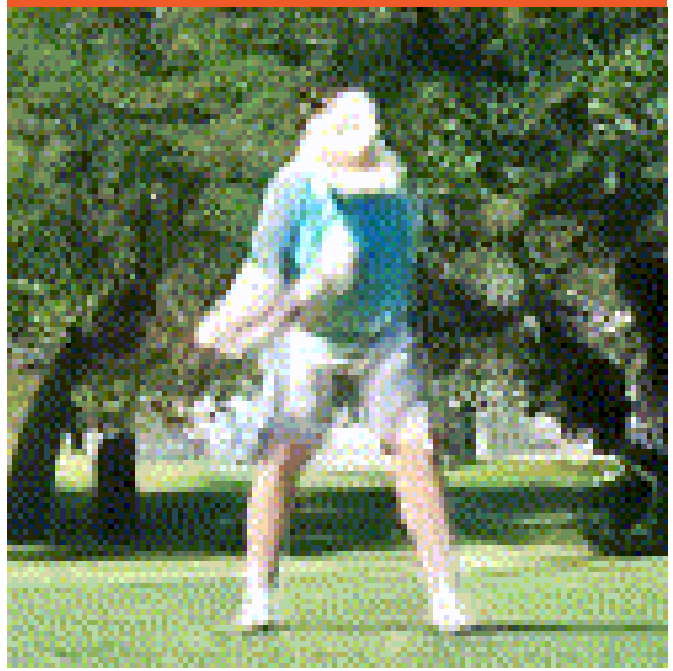


Video Prediction

Input



Output

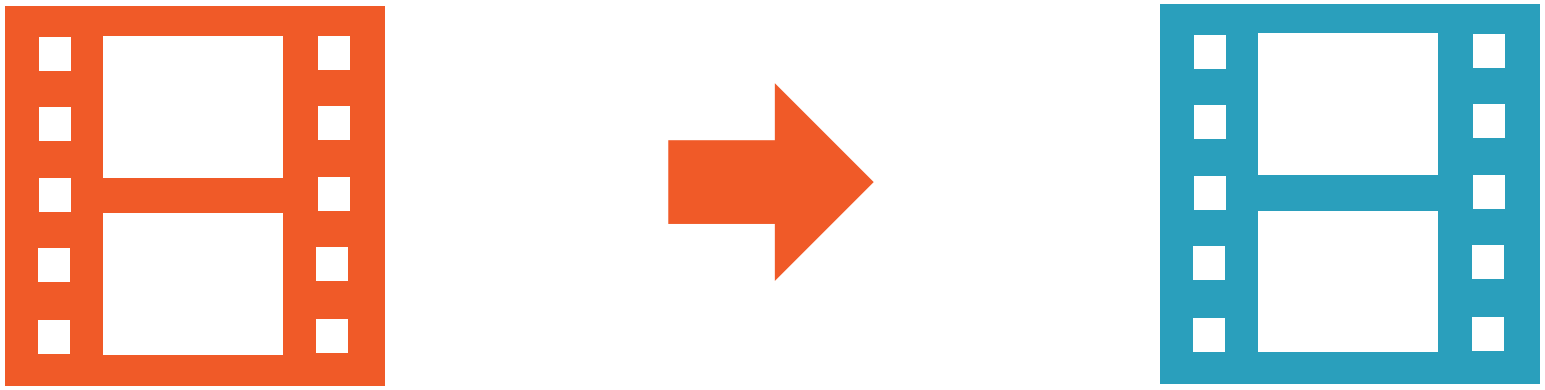


Source: https://sites.google.com/a/umich.edu/rubenevillegas/hierch_vid

Video Prediction

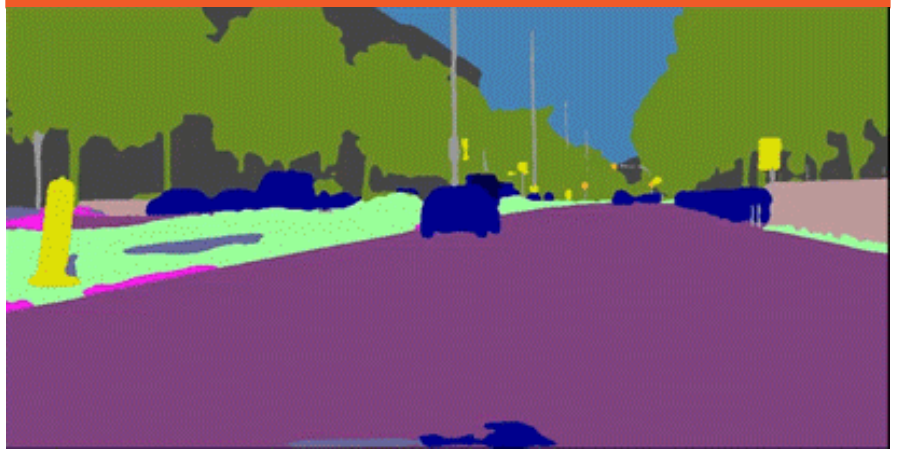
Future applications

Video Transfer



Video Transfer

Input



Output

Source: Nvidia Vid-to-Vid

Video Transfer

Future applications

Other Video Synthesis Tools

Video interpolation	Video prediction	Video transfer
Video completion	Video face synthesis	Video pose transfer

Video Completion

Input

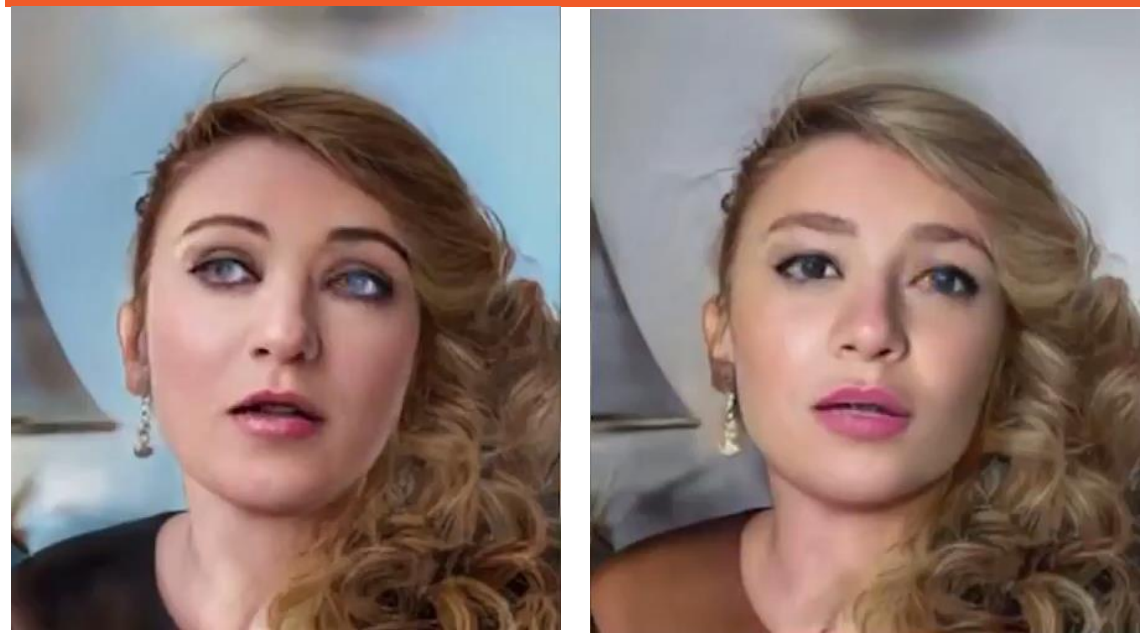
Output

Source: Deep Video Inpainting (2019)

Video Face Synthesis

Input

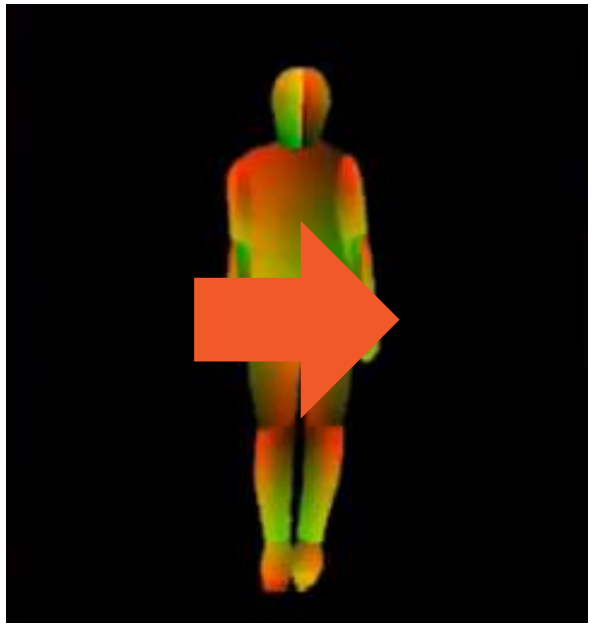
Output



Source: Nvidia Vid-to-Vid

Video Pose Transfer

Input

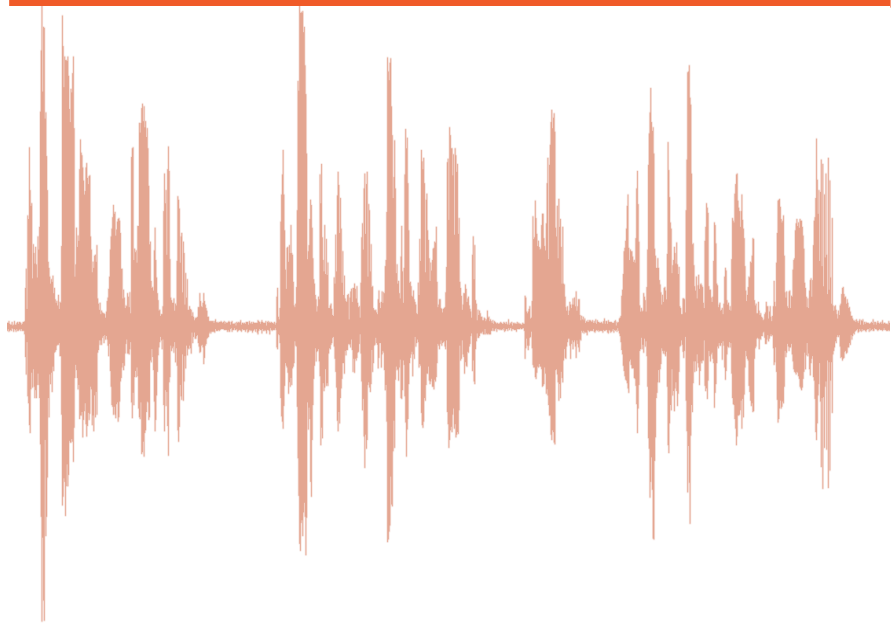


Output

Source: Nvidia Vid-to-Vid

Video Lip Syncing

Input



Output

Source: Synthesizing Obama: Learning Lip Sync from Audio

Demo

Demo Overview

Input

Output

No Action 100%

Teachable Machine

No Action

112 Pose Samples

Webcam Upload

Waving

133 Pose Samples

Webcam Upload

Thumbs Up

113 Pose Samples

Webcam Upload

Training

Model Trained

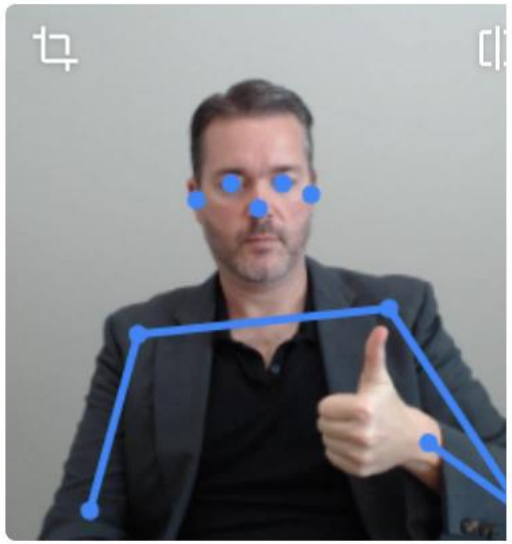
Advanced

Preview

Export Model

Input ON Webcam

HD Pro Webcam C920 (046d:082d)



Output

No Action

Wavi...

Thum... Up 100%

https://teachablemachine.withgoogle.com/

English (U.S.) release-2-3-1 - 2.3.1#3c7749 - 99

Summary

Video

Video Analysis

Video Synthesis

Demo